Basic Lighting Laws and Some Notes on Pole Spacing Geometry

We get quite a number of questions about some of the basic “laws” of lighting as they apply to outdoor lighting installations, as well as questions about adequate pole spacing for roadway lighting fixtures (semi-cutoff vs. full cutoff, for example). As these two issues relate to each other, this information sheet reviews a few of the basic facts. We will also give the definitions for a few of the basic terms.

Consider the light coming from a lighting source or a fixture, such as a street light, and falling on a surface such as the street pavement. The following definitions and lighting laws apply:

H = Mounting Height (MH) = the distance of the light above the surface.

D = distance from the source to where the light hits the surface.

Þ = angle between the vertical and the direction in question, that is, between the H and D lines. For example, 0° is straight down and 90° is horizontal.

X = the distance from the position directly beneath the light source and the point where the light hits the surface.

Note that X = H tan Þ, D = H / cos Þ, sin Þ = X / D, cos Þ = H / D, and tan Þ = X / H .

Inverse Square Law: The illumination **E** at a point on a surface varies directly with the luminous intensity **I** and inversely as the square of the distance **D** between the source and the point. If the surface is normal to the direction of the incident light, then $\mathbf{E = I / D^2}$. The light is getting spread out over a larger area as one gets further from the source.

Cosine Law, or *Lambert’s Law*: The illuminance on any surface varies as the cosine of the angle of incidence. The light is falling on a larger area than if it hit perpendicular to the surface. This can be combined with the inverse square law to get: $\mathbf{E = I \cos Þ / D^2}$ **.**

Cosine Cubed Law: By substituting H / cos Þ for D, the equation becomes: $\mathbf{E = I \cos^3 Þ / H^2}$**.**

Pole Spacing: The distance between two adjacent poles holding lighting fixtures. It is useful to consider the relative geometries involved. The angles and distances given in the following table could refer to either the cutoff angle (no light is emitted above this angle) or to the angle of maximum candlepower output of the luminaire. The optimum pole spacing is often considered to be the distance at which the maximum candlepower outputs from two luminaires meet on the ground. Other scenarios are possible, of course. The rows in the table have been calculated with either the angle or the value of X as the dependent variable, and the calculated values have been rounded off in most cases. The mounting height of the

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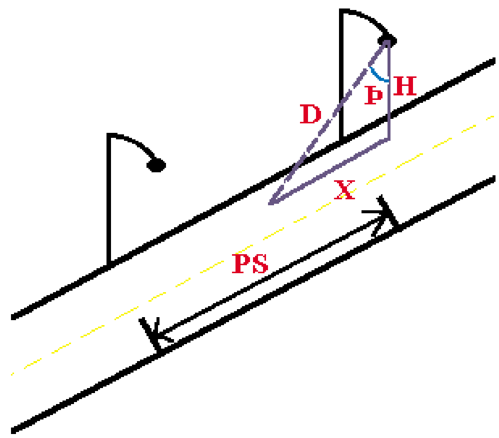
lighting luminaire above the ground, the horizontal spacing of one pole to the next, and the cutoff angle of the luminaire are all important issues in outdoor lighting design, just as much as is the choice of the luminaire, the lamp type, and the wattage. We give here a short table that relates some of the geometry of pole spacing relative to cutoff angle.

Angle (°)	X	D	D^2	$1 / D^2$	Pole Spacing If MH = 30 ft.
45	1.00	1.41	2.0	0.50	60
60	1.73	2.00	4.0	0.25	104
66	2.25	2.46	6.0	0.16	136
70	2.75	2.92	8.5	0.12	170
71.6	3.00	3.17	10	0.10	180
75.1	3.75	3.89	15	0.07	224
80	5.67	5.76	33	0.03	340
80.5	6.00	6.06	37	0.03	360
82.9	8.00	8.09	65	0.02	480
84.3	10.00	10.00	100	0.01	600
85	11.43	11.47	132	0.01	680
87.5	22.90	22.93	526	0.00	1370



At high cutoff angles, the X and D dimensions really stretch out, and the D^2 values show that there is little light left. There is no excuse for not having cutoff values be 80° to 82.5°. Any higher angles do not add anything to the effective light distribution, but they still do produce significant glare.

In fact, cutoff angles in the range 75° to 80° appear to make the best sense. This still allows sufficient overlap of the beams from two adjacent fixtures. The key to designing a “good” lighting fixture is to get the maximum light output at an angle of say 65° to 70°, thus getting a good light throw out away from under the light fixture (avoiding a “hot spot” under the fixture), while at the same time getting a sharp cutoff at an angle of 75° to 80°.

The result is then a nicely uniform distribution of light on the ground, out to a distance of at least six mounting heights from the pole, minimum glare, and no direct uplight. In addition, the designer must allow a good lighting distribution in an orthogonal direction (across the street, for example). It is possible, and such fixtures do exist. There is no excuse for not using them; they cost no more.

Units:

Candela (cd): A measure of luminous intensity (I). It is sometimes called candlepower, but that is not quite correct. It compares to radiant intensity, but holds only for the energy to which the eye is sensitive. It is a measure of radiant power, rather than energy, and it is weighted in terms of the eye sensitivity curve.

Lumen: The unit of luminous flux. A source emitting a luminous intensity of 1 cd uniformly in all directions will have a luminous flux of 1 lumen on a unit area of the sphere about it (the area of the sphere is 4π square units). An isotropic source of luminous intensity of 1 cd produces a total luminous flux of 4π lumens.

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Illuminance: The amount of light falling on a surface.

Lux: The unit of illuminance = the luminous flux per unit area of 1 square meter on a sphere of radius 1 meter.

Footcandle (fc): The unit of illuminance = the luminous flux per square foot on a sphere of radius 1 foot. One footcandle is approximately 10 lux.

Luminance: The amount of light reflected from a surface. It is sort of the “brightness” we see, i.e. the visual effect of the illuminance. It depends on the amount of illuminance and on the reflective properties of the surface as well as on the projected area on the plane perpendicular to the direction of view. The unit is candela per square meter (cd/m^2), or candela per square foot.